AI-BASED EYE DISEASE IDENTIFICATION & HEALTH RECOMMENDATION SYSTEM

BATCH NO: 33 | 23551A42B3 • 23551A4275 • 24555A4207

CLINICAL OPHTHALMOLOGY & EXPLAINABLE AI DIAGNOSTIC REPORT

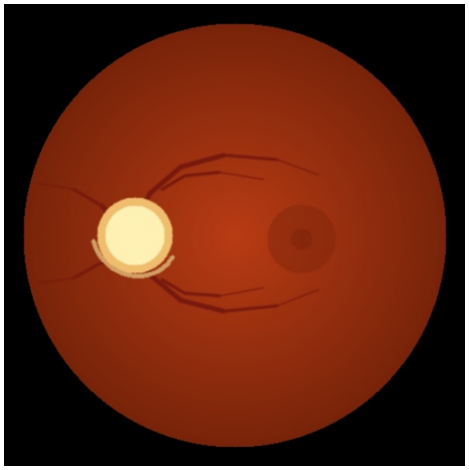
Patient Name: Jane Doe	**Patient ID:** PAT-TEST-99	**Exam Date:** 23 Sep 2026
Age / Gender: 52 yrs / Female	**Contact:** 9876543210	**Medical History:** Hypertension

1. ARTIFICIAL INTELLIGENCE DIAGNOSTIC SUMMARY

Detected Condition	Severity Staging	AI Confidence	Overall Risk Index
Glaucoma	**Moderate**	91.0%	62.0 / 100

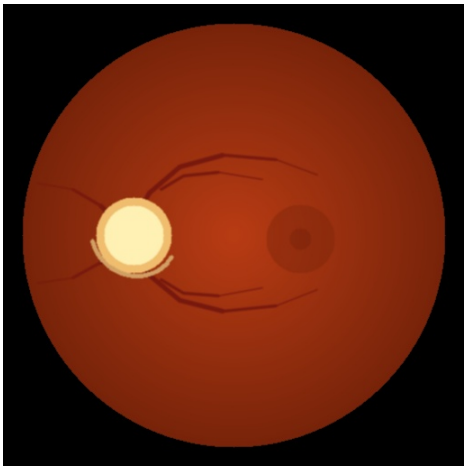
Cup-to-Disc Ratio (CDR): 0.68 **Macular Integrity:** 85.0% **Lesion Density:** 5.0% **Vessel Tortuosity:** 1.1

2. EXPLAINABLE AI (XAI) GRAD-CAM LESION LOCALIZATION



Original Retinal Fundus

Full vascular tree & optic disc



Grad-CAM Explainability Heatmap

Warm regions (Red/Yellow) highlight focal lesions

3. PERSONALIZED NUTRITIONAL & EYE CARE PRESCRIPTION

Target Nutrients:

• **Nitrate-Rich Leafy Greens**: Enhances nitric oxide bio-availability, improving ocular blood flow and trabecular meshwork drainage. (*Food sources: Kale, arugula, beet greens, spinach*)

• **Ginkgo Biloba & Flavonoids**: Neuroprotective effect on retinal ganglion cells and improves retrobulbar hemodynamics. (*Food sources: Dark chocolate (85%+), green tea, blueberries*)

• **Magnesium**: Relaxes retinal arteriolar vasospasm and optimizes ocular perfusion pressure. (*Food sources: Pumpkin seeds, almonds, black beans, dark leafy greens*)

Foods to Avoid:

• **Excessive Caffeine (>200mg/day)**: Can cause transient spikes in Intraocular Pressure (IOP).

• **Rapid High-Volume Fluid Ingestion**: Gulping more than 500ml of fluid in a short window spikes IOP.

Daily Eye Care Habits (20-20-20 Rule):
• Strict compliance with prescribed hypotensive eye drops at exact daily hours without missing doses.
• Avoid neckties tied too tightly and tight swimming goggles which raise episcleral venous pressure.

CLINICAL ACTION REQUIRED: Intermediate Stage - Clinical Attention Advised

Schedule a comprehensive in-person clinical exam within 2 to 4 weeks.

System Certification:

AI-assisted screening tool for clinical decision support. Final diagnostic confirmation must be validated by a registered ophthalmologist.

Consulting Ophthalmologist:

_____

Signature & Stamp